

what is prompt engineering ?

prompt engineering is the practice of designing and refining inputs (prompts) given to large language models (LLMs) or other AI systems so that they generate the most accurate, relevant, or useful outputs.

system prompt



system prompt is system instruction we need to a message how system need to behave

user prompt



user also have user prompt

Chaining

chaining means linking multiple "steps" of AI calls or functions together so the output of one step becomes the input for the next.

chain = prompt | model

chain

That's a chain: input → transform → search → summarize → final output.

what is output parser ?

In LangChain, an Output Parser is the piece that takes the raw text an LLM spits out and converts it into a structured format your program can actually use.

It is way to display your output

chain = prompt | model | Output parser

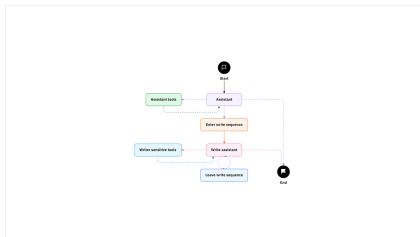
input first goes to prompt and next to model and then to outputparser that outputparser display the output from llm

Langchain vs Langgraph

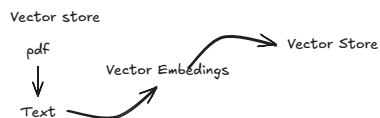
Langchain → Chaining → Sequential Execution of process

→ Semantic Search
→ Vector Databases
→ Text Splitter
→ Document Loader
Retrieve Summarize Answer i/o

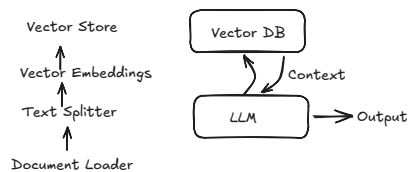
Langgraph: Multi Agents → Communicating Them selves → To solve Complex workflow



Memory in langgraph quite efficient



Traditional Rag



RAG : RAG (Retrieval-Augmented Generation) is an AI technique where the model first fetches info from external sources (retrieval) and then uses it to craft an answer (generation).